

FINAL PROJECT REPORT

Smartphone Usage, Digital Addiction & Well-Being Analytics

A Power BI Business Intelligence Project

Tools Used: MySQL, Power BI Desktop, Microsoft Excel

Dataset: 7,500 Smartphone Users

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1. Introduction

1.1 Project Overview

Smartphones have become deeply embedded in everyday life, shaping how people communicate, work, study, and relax. Alongside this convenience, growing screen time, constant notifications, and habitual app use have raised concerns about digital addiction and its effect on sleep, stress, and academic or work performance. This project, “Smartphone Usage, Digital Addiction & Well-Being Analytics,” analyses a dataset of 7,500 smartphone users to understand usage behaviour and its relationship with well-being indicators.

The end-to-end workflow covers data collection, cleaning and transformation in MySQL, and the design of an interactive Power BI dashboard and report that allow stakeholders — researchers, parents, educators, and platform designers — to explore usage patterns by age group, gender, stress level, and addiction level, and to draw data-backed conclusions about digital well-being.

1.2 Objectives

- To analyze smartphone usage and digital addiction patterns among 7,500 users and understand how usage affects daily life and well-being.
- To clean, transform, and model raw telemetry and survey data using MySQL and Power Query for reliable analysis.
- To identify relationships between screen time, sleep, stress, social media use, gaming, and academic/work impact.
- To design an interactive Power BI dashboard with KPIs, slicers, and drill-down visuals for dynamic exploration.
- To surface actionable insights that support digital well-being awareness for individuals and organizations.

2. Project and Initialization Planning Phase

2.1 Define Problem Statement

The problem statement is framed from the customer's point of view, describing who the users are, what they are trying to achieve, the barriers they face, why those barriers exist, and how the problem affects them.

I am	a smartphone user / digital product user
I'm trying to	understand and manage my smartphone usage
But	I do not clearly know which habits or factors are linked to smartphone addiction
Because	screen-time behavior, lifestyle habits, stress, and demographic factors may influence usage patterns
Which makes me feel	concerned and frustrated because targeted digital well-being support cannot be planned effectively

2.2 Project Proposal (Proposed Solution)

The proposed solution addresses the problem through a structured analytics pipeline that combines SQL-based data preparation with Power BI visual analytics.

Attribute	Details
Objective	Analyze smartphone usage and digital addiction patterns and understand how usage affects users' daily life and well-being.

Attribute	Details
Scope	Analyzing smartphone usage data, addiction levels, screen time, social media use, sleep, stress, and user demographics; presenting findings through a Power BI dashboard.
Problem Context	Many people spend a large amount of time on smartphones. High screen time, frequent app use, social media use, notifications, and other usage habits may be linked with digital addiction and changes in sleep, stress, and academic or work activities.
Description	The analysis helps identify common usage patterns and factors related to digital addiction, supporting better awareness of smartphone usage.
Approach / Impact	The dataset is cleaned and analyzed using MySQL and Power BI. SQL is used for data preparation and analysis; Power BI is used to create KPIs, charts, filters, and an interactive dashboard.
Key Features	Addiction analysis, screen-time analysis, age-group analysis, gender-wise analysis, stress and sleep analysis, social media usage analysis, and an interactive Power BI dashboard.

Resource Requirements

Resource Type	Description	Specification / Allocation
Computing Resources	CPU	Standard laptop / desktop processor
Memory	RAM	Minimum 8 GB recommended
Storage	Disk space	Space required for dataset, Power BI file, and project files
Frameworks	Data analysis tools	Power BI for dashboard development
Libraries	Data analysis libraries	Not required for the main dashboard; SQL and Power BI are used
Development Environment	Tools used	MySQL Workbench, Power BI Desktop, Excel
Data	Source, size, format	Smartphone usage dataset, 7,500 users, CSV format

2.3 Initial Project Planning

The project was planned using an agile sprint-based backlog covering data collection through validation.

Sprint	Functional Requirement	User Story	Task	Points	Priority
Sprint-1	Data Collection	USN-1	Collect and load smartphone usage data into the application for analysis.	2	High
Sprint-1	Data Preparation	USN-2	Clean and prepare the smartphone usage dataset by handling missing and inconsistent values.	2	High
Sprint-1	Data Understanding	USN-3	View important attributes such as age, gender, daily usage, app usage, and screen time.	1	Medium
Sprint-2	Dashboard	USN-4	View a dashboard showing key smartphone usage statistics and summary information.	3	High
Sprint-2	Usage Analysis	USN-5	Analyze smartphone usage patterns using charts and visualizations.	2	High
Sprint-2	User Analysis	USN-6	Compare smartphone usage across different user groups and categories.	2	Medium

Sprint	Functional Requirement	User Story	Task	Points	Priority
Sprint-3	Interactive Filtering	USN-7	Filter the dashboard using available fields to explore specific usage patterns.	2	Medium
Sprint-3	Insights	USN-8	Identify useful insights and trends from the smartphone usage data.	2	High
Sprint-3	Validation	USN-9	Verify that the dashboard visuals and calculated results correctly represent the source data.	1	High

Project focus: Power BI-based analysis of smartphone usage data, including data preparation, usage pattern discovery, dashboard visualization, interactive filtering, and insight generation, delivered across three sprints.

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

A structured data collection plan guided the sourcing and curation of the datasets used for the analysis, ensuring integrity and readiness for downstream SQL and Power BI processing.

Section	Description
Project Overview	Analyze smartphone usage and consumer behavior data — screen time, addiction indicators, and device/demographic attributes — to understand digital well-being trends and support awareness initiatives.
Data Collection Plan	Data was gathered from structured survey/telemetry exports and consolidated into a single relational table capturing usage metrics, behavioral indicators, and demographic fields.
Raw Data Sources Identified	The primary source is the Smartphone Usage dataset (7,500 records), supplemented by reference fields for device model, demographic segment, and behavioral classification.

Raw Data Sources

Source Name	Description	Format	Size	Access
Smartphone Usage & Behaviour Dataset	Core dataset of 7,500 users with screen time, app usage, sleep, stress, addiction level, and demographic fields.	CSV	~4 MB	Public
Device & Demographic Reference Data	Supplementary attributes for device model, OS, and user demographic segment used to enrich the fact table.	CSV / Excel	< 1 MB	Public

3.2 Data Quality Report

The following data quality log outlines the structural schema issues identified in the raw smartphone telemetry data, their severity, and the resolution steps applied.

Data Source	Data Quality Issue	Severity	Resolution Plan
Smartphone_Dataset	Missing feature columns: essential reporting metrics (e.g., brand tiers, usage categories) absent from the source table.	Moderate	Resolved by adding extra columns derived from original parent columns using conditional logic.
Smartphone_Dataset	Unusable composite parent fields: standalone parameters locked inside dense, multi-part text strings.	High	Created clear child fields from parent columns using string-splitting and text-extraction.

Data Source	Data Quality Issue	Severity	Resolution Plan
Smartphone_Dataset	Cryptic and complex system headers generated from database shortcodes, hindering visualization.	Moderate	Renamed columns to human-readable field headers.
Smartphone_Dataset	Vague attribute labels: overlapping structural names risked logic conflicts in relationship mapping.	Moderate	Remapped and standardized headers to establish strict field uniformity.

3.3 Data Exploration and Preprocessing

Exploration and preprocessing were carried out using SQL (MySQL) for validation/aggregation and Power Query for transformation, following the plan below.

Section	Description & Implementation
Data Overview	Comprehensive usage metrics across users: daily screen time, app opens, notifications, social media and gaming hours, sleep hours, stress level, academic/work impact, and addiction classification.
Data Cleaning	Checked for missing values (e.g., in age, daily_screen_time_hours, addiction_level) using a NULL-audit query; missing entries handled with averages or "Unknown" flags. Duplicate records were removed.
Data Transformation	Used Power Query to filter invalid/corrupt records, sort records chronologically where applicable, and create conditional calculated columns (e.g., addiction/stress categorization).
Data Type Conversion	Audited fields to fix mismatched types; numeric fields such as screen time and sleep hours cast to Decimal Number, categorical fields kept as Text.
Column Splitting & Merging	Split combined text fields into standalone attributes and merged related fields into unified, human-readable descriptor keys.
Data Modeling	Built a relational model with one-to-many relationships between lookup/dimension tables and the transactional usage fact table; DAX measures created for key metrics across demographics.
Save Processed Data	Exported the refined data model into Power BI (.pbix) for dashboard development, correlation analysis, and reporting.

Representative SQL queries used for exploration and validation (grouping by gender/addiction, computing averages, checking for nulls, and cross-tabulating addiction level with stress level and academic/work impact) are included in the Appendix (Section 10.1).

4. Data Visualization

4.1 Framing Business Questions

Visualization development transforms complex datasets into graphical formats that are easy to interpret, enabling stakeholders to gain insight and make informed decisions. Well-framed business questions guided the selection of metrics and the design of each visual in the Power BI report.

4.2 Developing Visualizations

Q1. Screen time variation by gender?	Q2. Users by addiction level?
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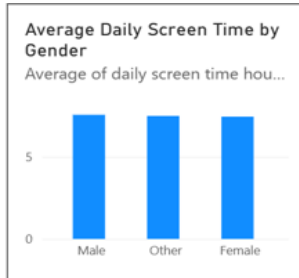


Fig 4.1 — Avg Daily Screen Time by Gender.

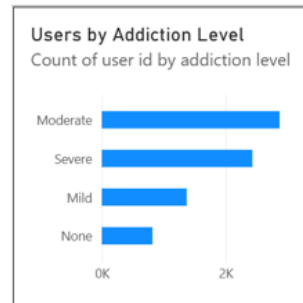


Fig 4.2 — Users by Addiction Level.

Q3. % of users flagged addicted?

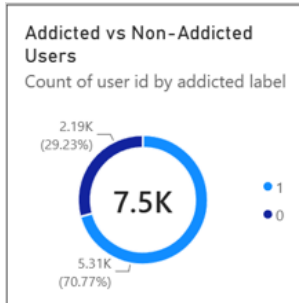


Fig 4.3 — Addicted vs Non-Addicted (29.23%).

Q4. Highest screen time by age group?

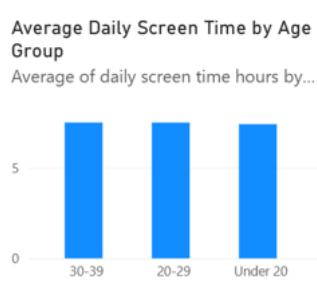


Fig 4.4 — Avg Screen Time by Age Group.

Q5. Screen time vs sleep correlation?

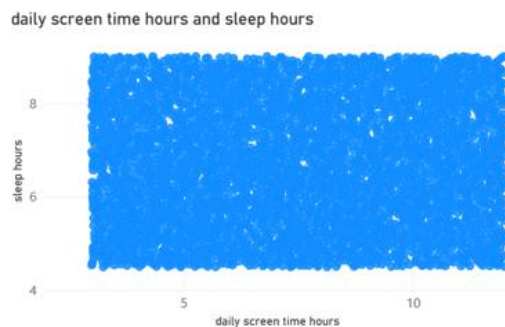


Fig 4.5 — Screen Time Hours vs Sleep Hours.

Q6. Gaming vs social media usage?



Fig 4.6 — Avg Gaming vs Social Media Usage.

Q7 & Q8. How can dashboard consumers dynamically filter the dataset by demographic and severity variables?

gender

☐ Female

☐ Male

☐ Other

Age Group

☐ 20-29

☐ 30-39

☐ Under 20

addiction level

☐ Mild

☐ Moderate

☐ None

☐ Severe

Fig 4.7 / 4.8 — Interactive vertical slicers for Gender / Age Group and for Addiction Level.

5. Dashboard

5.1 Dashboard Design Screenshot with Insights

The Power BI dashboard, “Digital Addiction & Screen Time Behavior Analysis,” combines KPI cards, categorical breakdowns, and a decomposition tree so stakeholders can drill from the full 7,500-user population down to specific demographic and behavioral segments. This canvas is also the basis for Report-1 in Section 6.1, where its insights are detailed.

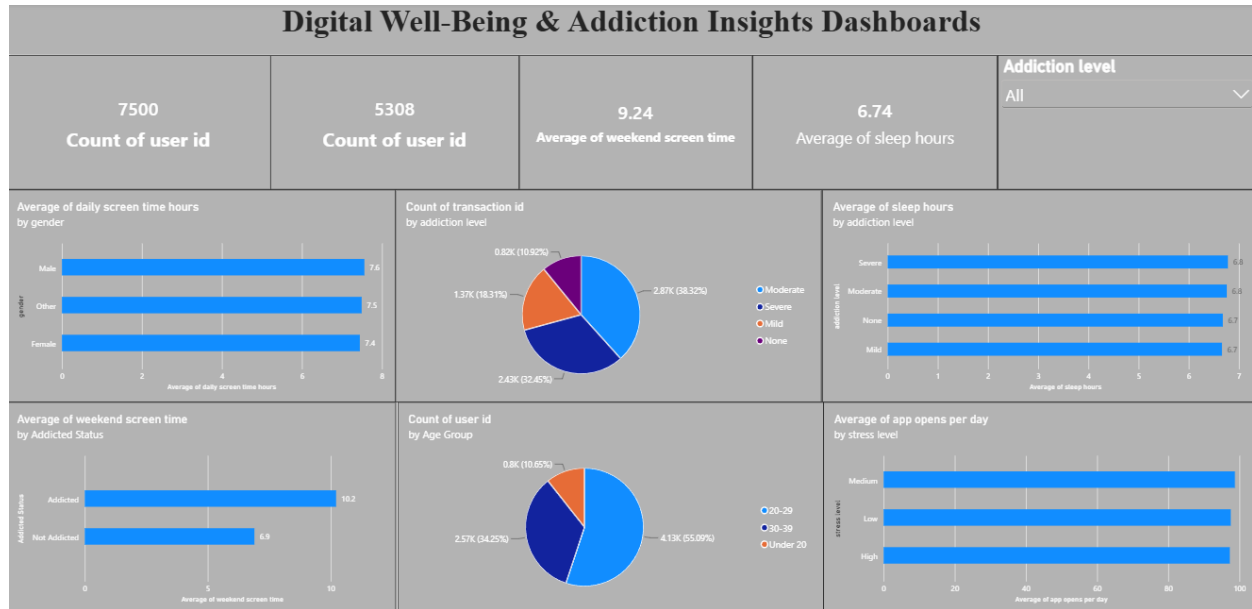


Fig 5.1 — Power BI Dashboard: Digital Addiction & Screen Time Behaviour Analysis.

- **KPI strip:** 97.83 average app opens/day, 3.27 average social media hours, 134.26 average notifications/day.
- **Drill-down:** A decomposition tree lets users break the 7,500-user count down by Age Group → Gender → Stress Level interactively.
- **Cross-filtering:** Addiction Level and Age Group slicers dynamically update every visual on the page for scenario exploration.

6. Report

6.1 Report Design Screenshot with Insights

Two report pages were designed in Power BI to present the smartphone usage findings from complementary angles — Report-1 focuses on behavioural and demographic patterns, while Report-2 focuses on well-being drivers and AI-assisted analysis.

Report – 1: Digital Addiction & Screen Time Behavior Analysis

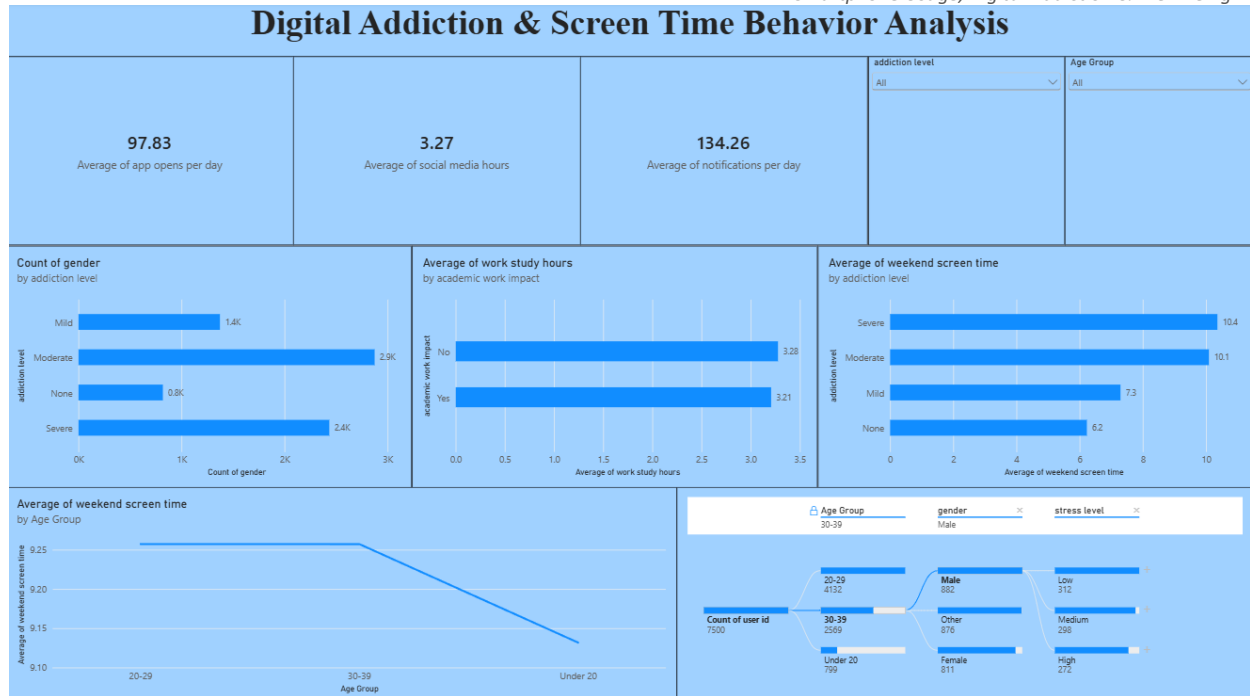


Fig 6.1 — Report-1: gender, addiction-level, and weekend screen-time breakdowns with a decomposition tree.

- **Core Digital Engagement:** Users average 97.83 app opens/day, 134.26 notifications/day, and 3.27 social media hours/day.
- **Prevalence of Addiction Levels:** Moderate is the largest group (2,874 users), followed by Severe (2,434 users).
- **Weekend Screen-Time Escalation:** Severe-addiction users average 10.4 weekend hours vs 6.2 hours for non-addicted users.
- **Demographic Breakdown Path:** Of 7,500 users, the 20-29 age group is largest (4,132); within 30-39/Male, Males = 882, Other = 876, Females = 211.

Report – 2: Smartphone Addiction & Digital Well-Being Insights

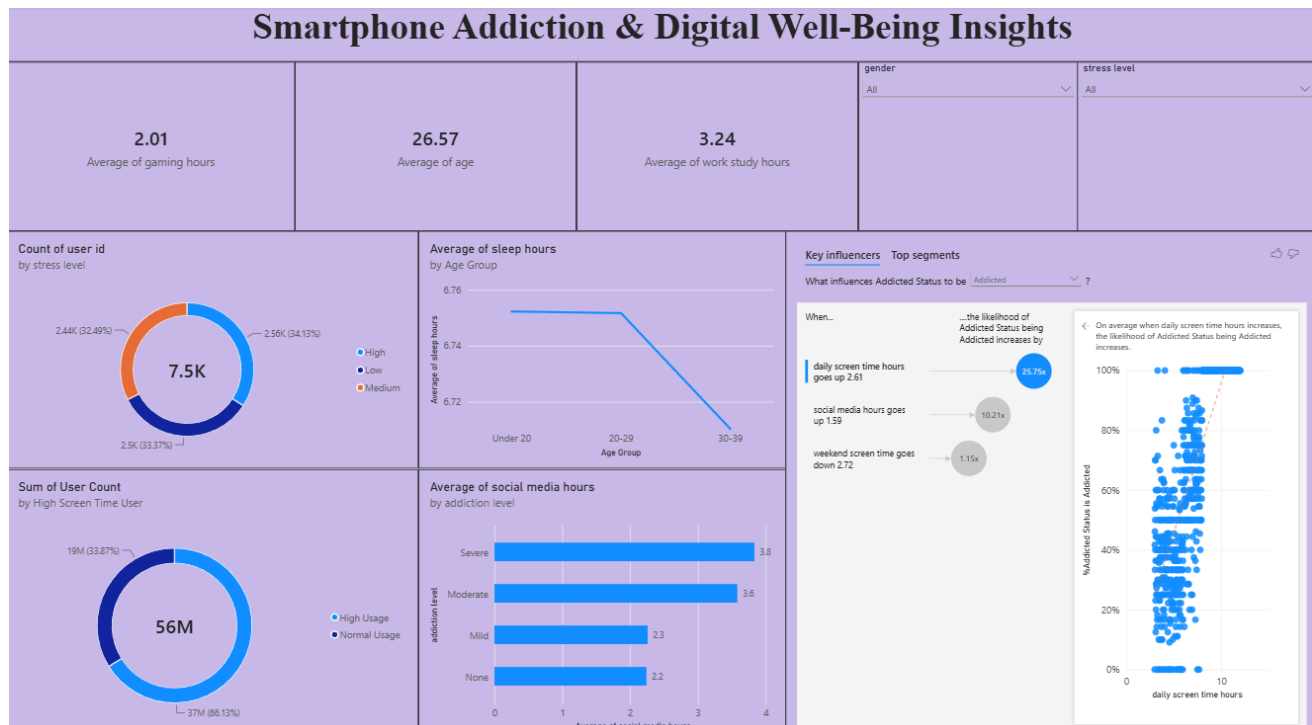


Fig 6.2 — Report-2: sleep, stress, and social-media breakdowns with a Key Influencers (AI) visual.

- **Stress Level Distribution:** High-stress cohort slightly leads at 34.13% (2.56K users) of the 7.5K population.

- **Sleep Deficit Trends:** The 30-39 age group has the lowest average sleep, roughly 6.71 hours, versus higher rest in younger cohorts.
- **Social Media Dependency:** Severe-addiction users log ~3.8 social media hours vs 2.2 hours for non-addicted users.
- **Key Behavioural Drivers (AI Insights):** Addicted-status likelihood rises 25.75x when daily screen time increases 2.61 hours, and 10.21x when social media hours rise 1.59.

7. Performance Testing

7.1 Utilization of Data Filters

Interactive slicers and cross-filtering were applied across both report pages so that every visual responds dynamically to the selected context.

- Gender and Age Group vertical checkbox slicers (dashboard page).
- Addiction Level checklist slicer (dashboard page) for isolating behavioral severities.
- Gender and Stress Level dropdown slicers (report/insights page).
- Drill-through decomposition tree filtering by Age Group, Gender, and Stress Level simultaneously.
- Cross-highlighting between KPI cards, bar charts, and the donut/scatter visuals on selection.

7.2 No. of Calculation Columns / Measures

DAX measures and calculated columns were created to support the KPI cards and category breakdowns shown on both report pages, including:

#	Measure / Calculated Column	Purpose
1	Average Daily Screen Time Hours	KPI and axis metric across gender, age group, and addiction level visuals.
2	Average Sleep Hours	Used in the sleep-vs-age-group trend line.
3	Average Social Media Hours	Used in the KPI card and social-media-by-addiction-level chart.
4	Average Gaming Hours	Used in the gaming-vs-social-media comparison chart.
5	Average App Opens / Notifications per Day	Core digital engagement KPI cards.
6	Addicted vs Non-Addicted % (Calculated Column)	Drives the donut chart percentage breakdown.
7	High Screen Time User Flag (Calculated Column)	Segments users for the Sum of User Count donut.
8	Average Weekend Screen Time	Used in the addiction-level and age-group weekend usage charts.

7.3 No. of Visualizations

Across the two Power BI report pages, 14 visuals were implemented in total — 6 KPI cards, 6 charts (bar, column, line, donut, scatter), 1 decomposition tree, 1 Key Influencers visual, and 5 interactive slicers, in addition to the 8 supporting visuals developed during the business-question phase described in Section 4.

8. Conclusion / Observation

The analysis of 7,500 smartphone users shows a clear, measurable relationship between smartphone usage intensity and digital well-being outcomes. Roughly 29.23% of users are flagged as digitally addicted, with the Moderate and Severe addiction levels together accounting for the majority of the tracked population. Higher daily and weekend screen time is consistently associated with reduced sleep, particularly in the 30-39 age group, and with elevated stress levels — the High-stress cohort represents just over a third of all users.

Social media and gaming engagement rise sharply with addiction severity: Severe-addiction users log nearly double the social media hours of non-addicted users. The Key Influencers analysis confirms that daily screen time and social media hours are the strongest predictors of addicted status, while a reduction in weekend screen time is associated with a lower likelihood of addiction. These findings validate the project's objective — the interactive Power BI dashboard successfully surfaces actionable, demographic-specific insights that can support digital well-being awareness initiatives.

9. Future Scope

- Incorporate time-series / longitudinal data to track how individual users' screen time and addiction levels change over months, not just a single snapshot.
- Extend the model with predictive analytics (e.g., a classification model for addiction risk) built on top of the existing Key Influencers findings.
- Add real-time data refresh from device telemetry APIs to keep the dashboard continuously updated.
- Expand demographic coverage (location, occupation, income) to enable deeper segmentation.
- Publish the dashboard to the Power BI Service with role-based row-level security for wider stakeholder access.

10. Appendix

10.1 Source Code

Representative MySQL queries used for data validation, exploration, and aggregation prior to Power BI modeling:

```
USE smartphone;
SELECT * FROM smartphone_usage;
-- Addiction count by gender
SELECT gender, COUNT(*), SUM(CASE WHEN addicted_label = 1 THEN 1 ELSE 0 END) AS addicted_count
FROM smartphone_usage GROUP BY gender;
-- Overall averages
SELECT AVG(daily_screen_time_hours), AVG(sleep_hours), AVG(stress_level = 'High')
FROM smartphone_usage;
-- Null / missing value audit
SELECT * FROM smartphone_usage
WHERE age IS NULL OR daily_screen_time_hours IS NULL OR addiction_level IS NULL;
-- Addiction label vs stress level cross-tab
SELECT addicted_label, stress_level, COUNT(*)
FROM smartphone_usage GROUP BY addicted_label, stress_level ORDER BY addicted_label, stress_level;
-- Addiction level summary
SELECT addiction_level, COUNT(*), AVG(daily_screen_time_hours), AVG(sleep_hours)
FROM smartphone_usage GROUP BY addiction_level;
-- Academic/work impact vs usage
SELECT academic_work_impact, AVG(daily_screen_time_hours), AVG(social_media_hours), AVG(gaming_hours)
FROM smartphone_usage GROUP BY academic_work_impact;
```

10.2 GitHub & Project Demo Link

Resource	Link
GitHub Repository	https://github.com/Neelima789/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics.git
Project Demo Video	https://drive.google.com/file/d/1GXEOH0GuABKRmi_I0Y-ps4V9GzrcD1h8/view?usp=sharing
Power BI File (.pbix)	smartphone.pbix